Matching crossing polynomials fail third-order q-log-convexity

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Abstract

For $F_n(x) = M_{2n}(x)$, the generating polynomial for the maximum number of mutually crossing arcs in a perfect matching, the third centered log-convexity iterate at index 5 has degree 34 and leading coefficient -14224. This disproves Chen's Conjecture 1.8. The input polynomials are already tabulated in the source and are independently verified by exhaustive enumeration.

Task 1. Full disproof

A perfect matching on the ordered set $[2n]$ has crossing number k if k is the maximum number of mutually crossing arcs in its standard arc diagram. In particular, two arcs (i, j) and (a, b) with $i < j$ and $a < b$ cross precisely when $i < a < j < b$ or $a < i < b < j$. Let

$$F_n(x) = M_{2n}(x) = \sum_k M_{2n,k} x^k.$$

We apply the source's centered operator across n:

$$(\mathcal{H}A)_n = A_{n-1}A_{n+1} - A_n^2.$$

Conjecture 1.8 of [1] asserts that every iterate is coefficientwise nonnegative, and proposes a further log-concavity property for those nonnegative iterates.

Theorem 1. *The matching polynomials satisfy*

$$(\mathcal{H}^3 F)_5(x) = -14224x^{34} + \textit{a polynomial of degree at most } 33. \tag{1}$$

They are therefore not infinitely q-log-convex.

Proof. The initial polynomials printed on p. 6 of [1] give

$$F_n(x) = x^n + a_n x^{n-1} + b_n x^{n-2} + \text{lower-degree terms},$$

with

n	2	3	4	5	6	7	8
a_n	2	9	20	35	54	77	104
b_n	0	5	70	315	891	2002	3900

. (2)

The unique matching with all n arcs mutually crossing has pairs $(1, n+1), \ldots, (n, 2n)$, accounting for the leading 1. The remaining entries in (2) are known source data; the complete independent enumeration certificate is described below.

Put $B_n = (\mathcal{H}F)_n$ and $C_n = (\mathcal{H}^2F)_n$. The leading terms of degree $2n$ cancel in B_n. For $3 \le n \le 7$, table (2) gives

$$a_{n-1} + a_{n+1} - 2a_n = 4,$$

so

$$B_n(x) = 4x^{2n-1} + v_n x^{2n-2} + \text{lower-degree terms},$$

where

$$v_n = b_{n-1} + b_{n+1} + a_{n-1}a_{n+1} - a_n^2 - 2b_n.$$

Substitution yields

n	3	4	5	6	7
v_n	19	95	186	314	474

. (3)

In $C_n = B_{n-1}B_{n+1} - B_n^2$, the terms of degree $4n-2$ cancel. The coefficient of the next power x^{4n-3} is

$$c_n = 4(v_{n-1} + v_{n+1} - 2v_n).$$

Hence

$$C_4 = 60x^{13} + \cdots, \qquad C_5 = 148x^{17} + \cdots, \qquad C_6 = 128x^{21} + \cdots. \tag{4}$$

It follows that $C_4C_6 - C_5^2$ has degree 34 and leading coefficient

$$60 \cdot 128 - 148^2 = 7680 - 21904 = -14224.$$

This proves (1) and refutes infinite q-log-convexity. □

This is an interior counterexample: only $F_2, \ldots, F_8$, or $M_4, \ldots, M_{16}$, are involved. At an ordinary positive parameter it gives the stronger numerical witness

$$(\mathcal{H}^3F)_5(10) = -1584481155042307779051353803951488000000000 < 0. \tag{5}$$

Independent counting certificate

The verifier enumerates all 2173623 matchings on the sets $[4], [6], \ldots, [16]$. At each recursive step it takes the smallest unmatched vertex i and pairs it with every remaining vertex j. Induction on the number of unmatched vertices shows that every matching occurs once and only once.

For each matching, it forms the graph whose vertices are its arcs and whose edges join pairs with alternating endpoints. Its maximum clique size is exactly the crossing number by definition. An exhaustive clique search therefore determines the entire coefficient row, not just the three coefficients used in (2). The row totals are $(2n-1)!!$, and every row agrees with the source. Independent hook-length sums over partitions with even row lengths give the same seven distributions.

Two exact polynomial multiplication methods reproduce all three iterates, including the cancellations in (3)–(4). Direct integer evaluation checks (5). No floating point or asymptotic approximation enters the verification. Review is by the authoring agent only.

The source's reported third-order positivity evidence is incompatible with (1), although its printed polynomials agree with the direct counts. The stronger regularity assertion on p. 7 explicitly starts with coefficientwise positivity, which already fails here. We do not claim that a quadratic log-concavity inequality must fail for the resulting signed coefficient list. Ordinary and second-order q-log-convexity, and log-concavity of each original coefficient row, remain separate questions.

Tasks 2–5. Follow-up directions

2. Decide second-order q-log-convexity of the matching crossing polynomials.

3. Determine all positive parameters x for which $(M_{2n}(x))_{n\geq 1}$ is infinitely log-convex.

4. Determine whether each fixed iterate is eventually coefficientwise nonnegative as n increases.

5. Derive formulas for the coefficients nearest the highest power in $\mathcal{H}^r F$, and classify the resulting sign changes.

References

[1] W. Y. C. Chen, *Log-concavity and q-Log-convexity Conjectures on the Longest Increasing Subsequences of Permutations*, arXiv:0806.3392v2, June 22, 2008. Conjecture 1.8 and pp. 5–7.

[2] MathDB, Problem 317480, accessed September 21, 2026.